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Trapeze hanger brace assembly

Abstract

A brace assembly can include a structural member including a sidewall; and a bracket including a baseplate defining a slot extending from an open end to a closed end, the baseplate defining at least one baseplate hinge hole at the open end; a clamp including a jaw member and a tightening member, the jaw member and the tightening member engaging the sidewall to couple the structural member to the bracket, the jaw member defining a clamp hinge hole; and a hinge pin engaging the at least one baseplate hinge hole and the clamp hinge hole.

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Background/Summary

REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. application Ser. No. 17/586,216, filed on Jan. 27, 2022, which is hereby specifically incorporated by reference herein in its entirety.

TECHNICAL FIELD
(1) This disclosure relates to trapeze pipe hangers. More specifically, this disclosure relates to seismic braces for trapeze pipe hangers.
BACKGROUND
(2) Pipes in buildings and other structures are commonly suspended below ceilings or roofs. One common form of pipe support is a trapeze hanger, which includes a crossmember suspended from two rods. The pipe can run between the two rods and be supported by the crossmember. Trapeze hangers can be unstable under lateral loading conditions. Seismic events, such as earthquakes, can impose lateral loads on trapeze hangers, which can result in damage to the hangers or the pipes supported by the hangers.
SUMMARY
(3) It is to be understood that this summary is not an extensive overview of the disclosure. This summary is exemplary and not restrictive, and it is intended to neither identify key or critical elements of the disclosure nor delineate the scope thereof. The sole purpose of this summary is to explain and exemplify certain concepts of the disclosure as an introduction to the following complete and extensive detailed description.
(4) Disclosed is a brace assembly including: a first bracket including: a first baseplate defining a first slot extending from a first open end to a first closed end, the first baseplate defining a baseplate hinge hole at the first open end and a first rounded end defining an axis at the first closed end; and a hinge pin engaging the baseplate hinge hole; a second bracket including a second baseplate defining a second rounded end aligned with the axis of the first rounded end; and a rod aligned with the axis.
(5) Also disclosed is a trapeze assembly including: a trapeze including: a crossmember; a rod coupled to the crossmember; and a fastener threadedly engaged with the rod; a first bracket including: a first baseplate defining a slot extending from an open end to a closed end, the first baseplate defining a baseplate hinge hole at the open end and a first rounded end defining an axis at the closed end; and a hinge pin engaging the baseplate hinge hole; and a second bracket including a second baseplate defining a second rounded end aligned with the axis of the first rounded end; wherein the rod is aligned with the axis.
(6) Further disclosed is an assembly including: a first baseplate defining a first slot extending from a first open end to a first closed end, the first baseplate defining a first baseplate hinge hole at the first open end, and a first rounded end defining a first axis at the first closed end; and a second baseplate defining a second slot extending from a second open end to a second closed end, the second baseplate defining a second baseplate hinge hole at the second open end, and a second rounded end defining a second axis at the second closed end; wherein the first axis is aligned with the second axis.
(7) Various implementations described in the present disclosure may include additional systems, methods, features, and advantages, which may not necessarily be expressly disclosed herein but will be apparent to one of ordinary skill in the art upon examination of the following detailed description and accompanying drawings. It is intended that all such systems, methods, features, and advantages be included within the present disclosure and protected by the accompanying claims. The features and advantages of such implementations may be realized and obtained by means of the systems, methods, features particularly pointed out in the appended claims. These and other

features will become more fully apparent from the following description and appended claims, or may be learned by the practice of such exemplary implementations as set forth hereinafter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The features and components of the following figures are illustrated to emphasize the general principles of the present disclosure. The drawings are not necessarily drawn to scale. Corresponding features and components throughout the figures may be designated by matching reference characters for the sake of consistency and clarity.

(2) FIG. **1** is a perspective view of a pipe trapeze assembly comprising a pipe trapeze and four brace assemblies in accordance with one aspect of the present disclosure.

(3) FIG. **2** is an exploded view of one of the brace assemblies of FIG. **1**.

(4) FIG. **3** is a cross-sectional view of another aspect of the pipe trapeze assembly, taken along line **3-3** shown in FIG. **1**, comprising two brace assemblies in accordance with another aspect of the present disclosure.

(5) FIG. **4** is a top perspective view of a baseplate of one of the brace assemblies of FIG. **1**.

(6) FIG. **5** is rear view of an open end of the baseplate of FIG. **4**.

(7) FIG. **6** is a detailed top perspective view of another aspect of the pipe trapeze assembly in accordance with another aspect of the present disclosure.

(8) FIG. **7** is a perspective view of another aspect of the pipe trapeze in accordance with another aspect of the present disclosure.

DETAILED DESCRIPTION

(9) The present disclosure can be understood more readily by reference to the following detailed description, examples, drawings, and claims, and the previous and following description. However, before the present devices, systems, and/or methods are disclosed and described, it is to be understood that this disclosure is not limited to the specific devices, systems, and/or methods disclosed unless otherwise specified, and, as such, can, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting.

(10) The following description is provided as an enabling teaching of the present devices, systems, and/or methods in its best, currently known aspect. To this end, those skilled in the relevant art will recognize and appreciate that many changes can be made to the various aspects of the present devices, systems, and/or methods described herein, while still obtaining the beneficial results of the present disclosure. It will also be apparent that some of the desired benefits of the present disclosure can be obtained by selecting some of the features of the present disclosure without utilizing other features. Accordingly, those who work in the art will recognize that many modifications and adaptations to the present disclosure are possible and can even be desirable in certain circumstances and are a part of the present disclosure. Thus, the following description is provided as illustrative of the principles of the present disclosure and not in limitation thereof.

(11) As used throughout, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “an element” can include two or more such elements unless the context indicates otherwise.

(12) Ranges can be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, another aspect includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another aspect. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint.

(13) For purposes of the current disclosure, a material property or dimension measuring about X or substantially X on a particular measurement scale measures within a range between X plus an industry-standard upper tolerance for the specified measurement and X minus an industry-standard lower tolerance for the specified measurement. Because tolerances can vary between different materials, processes and between different models, the tolerance for a particular measurement of a particular component can fall within a range of tolerances.
(14) As used herein, the terms “optional” or “optionally” mean that the subsequently described event or circumstance can or cannot occur, and that the description includes instances where said event or circumstance occurs and instances where it does not.
(15) The word “or” as used herein means any one member of a particular list and also includes any combination of members of that list. Further, one should note that conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain aspects include, while other aspects do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more particular aspects or that one or more particular aspects necessarily include logic for deciding, with or without user input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular aspect.
(16) Disclosed are components that can be used to perform the disclosed methods and systems. These and other components are disclosed herein, and it is understood that when combinations, subsets, interactions, groups, etc. of these components are disclosed that while specific reference of each various individual and collective combinations and permutation of these may not be explicitly disclosed, each is specifically contemplated and described herein, for all methods and systems. This applies to all aspects of this application including, but not limited to, steps in disclosed methods. Thus, if there are a variety of additional steps that can be performed it is understood that each of these additional steps can be performed with any specific aspect or combination of aspects of the disclosed methods.
(17) Disclosed is a pipe trapeze assembly and associated methods, systems, devices, and various apparatus. The pipe trapeze assembly can comprise a pipe trapeze and at least one brace assembly. It would be understood by one of skill in the art that the disclosed pipe trapeze assembly is described in but a few exemplary aspects among many. No particular terminology or description should be considered limiting on the disclosure or the scope of any claims issuing therefrom.
(18) FIG. **1** shows a perspective view of a pipe trapeze assembly **100** comprising a pipe trapeze **160** and at least one brace assembly **110**. In the aspect shown, the pipe trapeze assembly **100** can comprise four brace assemblies **110***a,b,c,d* (referred to generally as “the brace assembly **110**” or “the brace assemblies **110**”). Each brace assembly **110***a,b,c,d* can extend off the page.
(19) The pipe trapeze **160** can comprise a crossmember **162** and two rods **164***a,b*. Each of the rods **164***a,b* can extend off the page to mount to a structure, such as a ceiling, overhead beam, joist, or other suitable structure, for example and without limitation. In the aspect shown, the rods **164***a,b* can suspend the crossmember **162**. The rods **164***a,b* can extend in a substantially vertical orientation. The crossmember **162** can extend in a substantially horizontal orientation. The rods **164***a,b* can be substantially perpendicular to the crossmember **162**. The rods **164***a,b* can extend through openings **163** in the crossmember **162**, as demonstrated by the rod **164***a*. The openings **163** can be holes, slots, or any other type of opening. The crossmember **162** can comprise a channel, beam, bar, tube, angle, or other type of material stock for example and without limitation. The crossmember **162** can comprise a metal, such as iron, steel, aluminum, or any other suitable metal or alloy. In some aspects, the crossmember **162** can comprise a different material, such as a polymer, composite, or other suitable material. In the aspect shown, the crossmember **162** can be a piece of channel.
(20) As demonstrated by the brace assembly **110***d*, each brace assembly **110** can comprise a

structural member **112** and a bracket **114**. The bracket **114** can be coupled to the structural member **112**. In the present aspect, the structural member **112** can be a tubular member, such as a pipe or tube. In aspects utilizing tubes, the tubes can be round or other shapes, such as rectangular for example and without limitation. In some aspects, the structural member **112** can be a channel, beam, bar, angle, or other type or shape of material stock.

(21) The bracket **114** can comprise a clamp **120** and a baseplate **130**. The clamp **120** can comprise a jaw member **122** and a tightening member **124**. The tightening member **124** can be a fastener, such as a screw for example and without limitation. The tightening member **124** can threadedly engage with the jaw member **122**. The tightening member **124** can be tightened, or screwed into, the jaw member **122** to pinch a portion of the structural member **112** between the jaw member **122** and the tightening member **124**.

(22) In some aspects, the tightening member **124** can be designed to include a shear-off head **125**. The head of the tightening member **124**, such as a hex-head, can be designed to shear from the remainder of the tightening member **124** at a specified torque. The shear-off head **125** can simplify installation by providing feedback to an assembler that the tightening member **124** has been sufficiently tightened without requiring a calibrated tool, such as a torque wrench. The shear-off head **125** can also prevent an assembler from overtightening the tightening member **124**. Because the head shears off, tampering with the tightening member **124** can be prevented, such to stop a person from loosening the tightening member **124**. Additionally, the absence of the head on the tightening member **124** can provide quick visual confirmation that the tightening member **124** has been properly tightened. For example, a worker standing on the floor can look upwards towards brace assemblies **110** suspended from a ceiling and visually confirm that the tightening member **124** has been properly tightened based on the absence of the shear-off head **125** without requiring access to the tightening member **124**, such as by climbing a ladder.

(23) The clamp **120** can be hingedly coupled to the baseplate **130** by a hinge pin **116**. The hinge pin **116** can be a fastener, such as a bolt, screw, rivet, pin, stud, nut, or other suitable type of fastener or combinations thereof. In the aspect shown, the hinge pin **116** can be a screw, which can threadedly engage the baseplate **130**. The baseplate **130** of each brace assembly **110** can fit around one of the rods **164***a,b*. As demonstrated by brace assemblies **110***c,d* and the rod **164***b*, multiple baseplates **130** can be fit around each rod **164***a,b*, and the baseplates **130** can be stacked atop one another.

(24) The pipe trapeze **160** can further comprise reinforcement plates, or backing plates, **166** and fasteners **168**. The backing plates **166** can be positioned in contact with both a top side **170** and a bottom side **172** of the crossmember **162**. The fasteners **168** can be positioned both above and below (fasteners **168** shown below the crossmember **162** and backing plates **166** in FIG. **3**) the backing plates **166**. The crossmember **162** can be positioned between the backing plates **166**, and the backing plates **166** can be positioned between the fasteners **168** of a given rod **164***a,b*. In the present aspect, the rods **164***a,b* can be at least partially threaded, such as with external threading **165** for example and without limitation, and the fasteners **168** can be nuts, which can threadedly engage with the rods **164***a,b*. The fasteners **168** can be tightened towards one another on the respective rod **164***a,b* to clamp the baseplates **130** to the top backing plate **166** and the backing plates **166** to the top side **170** and the bottom side **172** of the crossmember **162**.

(25) A lateral direction **197** can be defined in a direction extending horizontally between the rods **164***a,b*. A length L of the crossmember **162** can be oriented parallel to the lateral direction **197**. A longitudinal direction **198** can be defined extending horizontally in a direction perpendicular to the lateral direction **197** and the length L of the crossmember **162**. A vertical direction **199** can be perpendicular to each of the lateral direction **197** and the longitudinal direction **198** and aligned with the acting direction of the force of gravity.

(26) Each brace assembly **110** can be rotationally indexed around the respective rods **164***a,b* prior to tightening the fasteners **168**. Tightening the fasteners **168** can secure the brace assemblies **110** so that the respective brace assemblies **110** are rotationally fixed about the respective rods **164***a,b*. As

described below in greater detail, the baseplates **130** can interlock when stacked to secure the brace assemblies **110** attached to the same rod **164***a,b* in a set orientation relative to one another. In the aspect shown, the structural members **112** of brace assemblies **110***a,d* can extend laterally outward and vertically upward from the crossmember **162**, with respect to the lateral direction **197** and the vertical direction **199**. The structural members **112** of brace assemblies **110***b,c* can extend longitudinally outward and vertically upward from the crossmember **162**, with respect to the longitudinal direction **198** and the vertical direction **199**.

(27) The hinge pins **116** can also provide a range of angles of the structural members **112** and the clamps **120** relative to the respective baseplates **130**. In some aspects, the structural members **112** and the clamps **120** can rotate over 180-degrees relative to the respective baseplates **130**. For example and without limitation, the structural members **112** can be horizontally oriented, as shown in FIG. **3**, vertically oriented, as shown in FIG. **7**, or diagonally oriented as shown in FIGS. **1** and **6**. In some aspects, the hinge pin **116** can be tightened to secure the structural members **112** and the clamps **120** at a specific angle relative to the baseplate **130**. In some aspects, the hinge pin **116** can be a screw that can threadedly engage the baseplate **130**, and the screw can be tightened to the baseplate **130**. In some aspects, the hinge pin **116** can be a nut and bolt, which can be tightened together.

(28) FIG. **2** is an exploded view of one of the brace assemblies **110** of FIG. **1**. The structural member **112** can comprise a sidewall **212**, which can define a bore **214** extending through the structural member **112**. Specifically, the bore **214** can be defined by an inner surface **216** of the structural member **112**, and the structural member **112** can define an outer surface **218** on a opposite side of the sidewall **212**. An end **220** of the structural member **112** can define an opening **222** to the bore **214**.

(29) The jaw member **122** of the clamp **120** can define a jaw **240**, a tightening arm **242**, and a base **244**. The jaw **240** and the tightening arm **242** can each be connected to and protrude from the base **244**. The jaw **240** and the tightening arm **242** can extend in substantially the same direction from the base **244**. A hinge end **252** of the jaw member **122** can be defined opposite from the jaw **240** and the tightening arm **242**. In some aspects, at least one surface of the base **244** can be parallel to at least one surface of the tightening arm **242**. A gap **246** can be defined between the jaw **240** and the tightening arm **242**.

(30) The tightening member **124** can extend through the tightening arm **242**. The tightening member **124** can threadedly engage with the tightening arm **242**. The tightening member **124** can be rotated to drive a tip **248** of the tightening member **124** into the gap **246** and towards the jaw **240** or out from the gap **246** and away from the jaw **240**. The hinge end **252** of the base **244** can define a clamp hinge hole **250**, which can be sized to receive the hinge pin **116**.

(31) The baseplate **130** can comprise two legs **264***a,b* connected by a closed end **262**. An open end **260** can be defined opposite from the closed end **262** by the legs **264***a,b*. The baseplate **130** can define a U-shape, or a horseshoe shape, for example and without limitation. A slot **265** can be defined between the legs **264***a,b* extending from the open end **260** towards the closed end **262**.

(32) The closed end **262** and portions of the legs **264***a,b* adjacent to the closed end **262** can define an upper surface **266**. The legs **264***a,b* can respectively define a pair of cars **266***a,b* at the open end **260**. The ears **266***a,b* can extend upwards from the upper surface **266**. The hinge end **252** of the base **244** can be sized to fit between the ears **266***a,b* and at least partially into the slot **265** through the open end **260**.

(33) The cars **266***a,b* can each define a baseplate hinge hole **268***a,b*, respectively. The baseplate hinge hole **268***a* can be a smooth hole. The baseplate hinge hole **268***b* can be a threaded hole. In some aspects, the baseplate hinge hole **268***a* can be larger in diameter than the baseplate hinge hole **268***b*. In some aspects, both hinge holes **268***a,b* can be smooth holes.

(34) The hinge pin **116** can be inserted through the baseplate hinge hole **268***a*, the clamp hinge hole **250**, and a threaded portion **270** of the hinge pin **116** can threadedly engage with internal threading

269 of the baseplate hinge hole **268***b* to hingedly couple the clamp **120** to the baseplate **130**. In some aspects, tightening the hinge pin **116** can pinch the cars **266***a*,*b* closer together, and the baseplate **130** can grip the hinge end **252** of the base **244** to secure the clamp **120** at a set angle relative to the baseplate **130**, thereby preventing rotation of the clamp **120** relative to the baseplate **130** about the hinge pin **116**.
(35) FIG. **3** is a cross-sectional view of another aspect of the pipe trapeze assembly **100**, taken along line **3-3** of FIG. **1**, comprising the pipe trapeze **160** of FIG. **1** and two brace assemblies **110***a*,*b*, in accordance with another aspect of the present disclosure.
(36) In the aspect shown, each clamp **120** can be pivoted about the respective hinge pin **116** relative to the respective baseplate **130** so that the structural members **112** extend horizontally outwards without extending upwards or downwards relative to the vertical direction **199**. In the aspect shown, the structural members **112** can extend outwards in the lateral direction **197**.
(37) The fasteners **168** can couple the baseplates **130** to the crossmember **162**. As demonstrated by the rod **164***b*, the fasteners **168** can define internal threading **368**, which can threadedly engage with the external threading **165** of each rod **164***a*,*b*. In the aspect shown, the rod **164***b* can define the external threading **165** along its entire length, for example and without limitation. Specifically, the fasteners **168** can secure the baseplate **130** to the top backing plate **166** and secure the backing plates **166** to the top and bottom of the crossmember **162**. As demonstrated by the brace assembly **110***a* and the rod **164***a*, each rod **164***a*,*b* can extend through the slot **265** (shown bisected in the present view).
(38) One or more runs of piping **399** can be supported on the crossmember **162**, which in turn can be suspended by the rods **164***a*,*b*. The piping **399** can extend in the longitudinal direction, which can extend into and out from the page with respect to the present viewing angle (longitudinal direction **198** also shown in FIG. **1**).
(39) As demonstrated by the clamp **120** of the brace assembly **110***b*, the clamp **120** can grip a portion of the sidewall **212** of the structural member **112**. The jaw **240** can extend through the opening **222** (shown in FIG. **2**) and into the bore **214**, and the end **220** can abut the base **244** of the jaw member **122**. The jaw **240** can engage with the inner surface **216** of the structural member **112**. The tip **248** of the tightening member **124** can engage with the outer surface **218** of the structural member **112**. In some aspects, the tip **248** can penetrate or deform the sidewall **212**.
(40) The tightening member **124** can define external threading **324**, which can threadedly engage with internal threading **342** within a bore **344** of the tightening arm **242**. By rotating the tightening member **124**, the tip **248** can be driven into the gap **264** until the tip **248** engages with the sidewall **212** and pinches the sidewall **212** between the jaw **240** and the tip **248**.
(41) As similarly noted above, in some aspects, the structural member **112** may not be a tubular member. In aspects wherein the structural member **112** defines a different shape, the jaw **240** and the tip **248** of the tightening member **124** can engage with a different part of the structural member **112**, such as a web, flange, or sidewall portion as seen in beams, angled shapes, channels, bars of various profiles, and other shaped stock, for example and without limitation.
(42) FIG. **4** is a top perspective view of the baseplate **130** of FIG. **1**. The slot **265** can define a slot end **464** where the slot **265** terminates at the closed end **262**. The upper surface **266** can be substantially planar. A countersink **460** can be recessed below the upper surface **266** at the slot end **464**. The slot end **464** can define a rounded end **465** with an axis **468**. In some aspects, the rounded end **465** can define a frustoconical portion **466**. In some aspects, the rounded end **465** can define a cylindrical portion. The countersink **460** can be coaxially centered around the rounded end **465** about the axis **468**. The countersink **460** can be substantially circular in shape.
(43) FIG. **5** is a rear view of the open end **260** of the baseplate **130** of FIG. **1**. The baseplate **130** can define a lower surface **566** opposite from the upper surface **266**. The lower surface **566** can be substantially parallel to the upper surface **266**. The slot **265** can extend through the baseplate **130** from the upper surface **266** to the lower surface **566**. The slot **265** can define a pair of side surfaces

565*a*,*b* extending from the open end **260** to the rounded end **465**. The side surface **565***a* can be defined by the leg **264***a*, and the side surface **565***b* can be defined by the leg **264***b*. The side surfaces **565***a*,*b* can slope towards each other from the upper surface **266** to the lower surface **566**, such that the slot **265** can be narrower where the slot **265** meets the lower surface **566** compared to where the slot **265** meets the upper surface **266**. The slot **265** can narrow from the upper surface **266** towards the lower surface **566**. As shown, the cars **266***a*,*b* can taper, or become narrower, as the cars **266***a*,*b* extend away from the lower surface **566**.
(44) FIG. **6** is a detailed top perspective view of the pipe trapeze assembly **100** showing the brace assemblies **110***a*,*b*, the rod **164***a*, and the crossmember **162** of the pipe trapeze **160**.
(45) The brace assemblies **110***a*,*b* are shown in another rotational orientation relative to the crossmember **162**. In the orientation shown, each of the structural members **112** of the brace assemblies **110***a*,*b* can extend both laterally and longitudinally outward from the crossmember **162**, with respect to the lateral direction **197** and the longitudinal direction **198**. The structural members **112** can also extend upwards from the crossmember **162** with respect to the vertical direction **199**.
(46) With respect to FIG. **6**, a baseplate **130***a* can refer to the baseplate of the brace assembly **110***a*, and a baseplate **130***b* can refer to the baseplate of the brace assembly **110***b*. As demonstrated by the baseplate **130***b*, the fastener **168** can sit in the countersink **460**, which can secure the rod **164***a* relative to the baseplate **130***b*. For example and without limitation, engagement between the fastener **168** and the countersink **460** can prevent the rod **164***a* from sliding along the slot **265** towards the open end **260**. The rod **164***a* can be secured at the rounded end **465** (shown in FIG. **4**), which can be shaped to closely fit the rod **164***a*.
(47) The baseplates **130***a*,*b* can be shaped and sized so that the rounded ends **465** can be aligned together when the baseplates **130***a*,*b* are stacked on top of each other. For example, the axis **468** (shown in FIG. **4**) of the respective rounded ends **465** can be aligned together as well as with the rod **164***a*. As similarly described above, engagement between the fastener **168** and the countersink **460** can secure the rod **164***a* at the rounded end **465** of the baseplate **130***b*. The sizing of the baseplates **130***a*,*b* can cooperate to secure the rod **164***a* at the rounded end **465** of the baseplate **130***a*.
(48) Specifically, the leg **264***b* of the baseplate **130***b* can engage with the cars **266***a*,*b* of the baseplate **130***a* to secure the rod **164***a* at the rounded end **465** of the baseplate **130***a*. For example, a width of the leg **264***b* can be sized to space the rod **164***a* sufficiently far from the cars **266***a*,*b* to position and secure the rod **164***a* at the rounded end **465** of the baseplate **130***a*. The shape and sizing of the leg **264***b* and the cars **266***a*,*b* can also be configured to interlock the baseplates together, thereby preventing relative rotation of the baseplates **130***a*,*b* relative to one another and securing the baseplates **130***a*,*b* in a set orientation relative to one another. In the aspect shown, the set orientation can be a perpendicular orientation relative to one another. For example and without limitation, the slots **265** of the respective baseplates **130***a*,*b* can be offset 90-degrees from one another.
(49) Similarly, in some aspects, the closed end **262** of the top baseplate **130***b* can be positioned against the cars **266***a*,*b* of the bottom baseplate **130***a* to interlock the baseplates **130***a*,*b* together, thereby preventing rotation of the baseplates **130***a*,*b* relative to one another. The shape and size of cars **266***a*,*b* and the closed end **262** can be configured to interlock and secure the baseplates **130***a*,*b* in another set orientation, such as a parallel orientation wherein the slots **265** can be 180-degree offset from one another. The closed end **262** and the cars **266***a*,*b* can be sized to secure the rod **164***a* at the rounded end **465** of each baseplate **130***a*,*b* when stacked in the 180-degree offset orientation. For example, a width of the closed end **262** can be sized to space the rod **164***a* sufficiently far from the cars **266***a*,*b* to position and secure the rod **164***a* at the rounded end **465** of the lower baseplate **130***a*.
(50) In common applications, the rods **164***a*,*b* (rod **164***b* shown in FIG. **1**) of the pipe trapeze **160** can primarily support, such as by suspending, the crossmember **162** and any pipes supported by it,

such as the piping **399** (shown in FIG. **3**), in the vertical direction **199**. Due to their length and connection mechanism, the rods **164***a,b* typically offer high flexibility in the lateral direction **197** and particularly the longitudinal direction **198**, and pipe trapezes **160** are frequently utilized where longitudinal expansion of piping is expected, such as piping runs experiencing substantial variations in pressure and/or temperature, for example and without limitation.

(51) In the occurrence of a seismic event, the crossmember **162** and the piping supported by it may move excessively if unbraced. The brace assemblies **110***a,b,c,d* can provide support in the lateral direction **197** and/or the longitudinal direction **198**. The brace assemblies **110***a,b,c,d* (brace assemblies **110***c,d* shown in FIG. **1**) can offer the greatest support in directions parallel with or at least partially aligned with the structural member **112** of each brace assembly **110***a,b,c,d*. For example and without limitation, a brace assembly **110** oriented with a structural member **112** extending laterally outward from the crossmember **162** or laterally outward and vertically upward from the crossmember **162** can provide the greatest support in the lateral direction **197** but limited support in the longitudinal direction **198**. A brace assembly **110** oriented with a structural member **112** extending longitudinally outward from the crossmember **162** or longitudinally outward and vertically upward from the crossmember **162** can provide the greatest support in the longitudinal direction **198** but limited support in the lateral direction **197**. When the pipe trapeze **160** deflects horizontally, the tapered shape of the slot **265** (shown in FIG. **5**) and the frustoconical portion **466** (shown in FIG. **4**) can cooperate to provide clearance for the rods **164***a,b* to deflect from the vertical direction relative to the baseplates **130** without the brace assemblies **110** exerting bending moments on the rods **164***a,b*, which could potentially damage the rods **164***a,b*.

(52) In some applications, one or more brace assemblies **110***a,b* can be retrofitted to an existing pipe trapeze **160**. For example and without limitation, the top, or upper, fastener **168** can be rotated up the respective rod **164***a,b* and away from the crossmember **162** to provide clearance to slip one or more baseplates **130** over the respective rod **164***a,b*. The fastener **168** can be tightened against the one or more baseplates **130***a,b* to secure the orientation of each baseplate **130***a,b* relative to the crossmember **162**. Hinge pins **116** can be inserted to secure the baseplates **130***a,b* to the respective clamps **120** and structural members **112** (shown in FIG. **1**) to assemble the brace assemblies **110**. Similarly, additional brace assemblies **110***a,b* can be added to or removed from the existing pipe trapeze **160** without disassembling the pipe trapeze **160**. For example and without limitation, one, two, three, four, or more brace assemblies **110** can be secured to the same rod **164**.

(53) FIG. **7** shows another aspect of a pipe trapeze **760**. In the aspect shown, the crossmember **162** can be supported in the vertical direction **199** by the brace assemblies **110***a,b*. The rods **164***a,b* and fasteners **168** can couple the brace assemblies **110***a,b* to the crossmember **162**. The brace assemblies **110***a,b* can extend upwards and off the page, where the brace assemblies **110***a,b* can mount to a structure, such as a ceiling, overhead beam, joist, or other suitable structure, for example and without limitation. In the aspect shown, the rods **164***a,b* do not provide vertical support to the crossmember **162**. In some aspects, the rods **164***a,b* can be bolts, rivets, screws, or any other suitable fasteners or combinations thereof.

(54) In the orientation shown, the baseplates **130** can offset the structural members **112** in the longitudinal direction **198**, relative to the crossmember **162** and the rods **164***a,b*. For example, the open ends **260** of the baseplates **130** can face in the longitudinal direction **198**, which can position the hinge pins **116** and structural members **112** longitudinally outward from the crossmember **162** and the rods **164***a,b*. In some aspects, the baseplates **130** can offset the structural members **112** in the lateral direction **197**, relative to the crossmember **162** and the rods **164***a,b*. As one example, the open ends **260** of the baseplates **130** can face away from one another, which can position the hinge pins **116** and the structural members **112** laterally outward from the crossmember **162** and the rods **164***a,b*. As a second example, the open ends **260** of the baseplates **130** can be positioned facing one another, which can position the hinge pins **116** and the structural members **112** laterally inward from the rods **164***a,b*.

(55) In the aspect shown, the structural members **112** are shown extending upwards from the crossmember **162** substantially parallel to the vertical direction **199**. In some aspects, the structural members **112** can support the crossmember **162** while extending vertically upward as well as longitudinally outward, laterally outward, or both longitudinally and laterally outward, relative to the lateral direction **197**, the longitudinal direction **198**, and the vertical direction **199**. In some aspects, the pipe trapeze **760** can comprise more than two brace assemblies **110***a,b*.
(56) Various methods for using the brace assembly **110** to brace the pipe trapeze **160** are contemplated. In some aspects, a method for bracing a pipe trapeze with a brace assembly can comprise sliding a slot of a baseplate of the brace assembly onto a rod of the pipe trapeze through an open end of the baseplate until the rod is positioned at a slot end of the slot, the rod fixed to and suspending a crossmember of the pipe trapeze, the slot extending from the open end of the baseplate towards a closed end of the baseplate, the slot end defined where the slot terminates at the closed end; coupling the baseplate to a structural member of the brace assembly; and tightening a fastener against the baseplate to secure the baseplate relative to the crossmember and the rod, the fastener threadedly engaged with the rod.
(57) In some aspects, tightening the fastener against the baseplate can comprise positioning the fastener in a countersink of the baseplate, and the countersink can be centered around the slot end. In some aspects, coupling the baseplate to the structural member can comprise inserting a hinge pin through the baseplate and through a clamp of the brace assembly to hingedly couple the baseplate to the clamp, and the clamp can be coupled to the structural member. In some aspects, inserting the hinge pin through the baseplate and through the clamp to hingedly couple the baseplate to the clamp can comprise: inserting the hinge pin through a first baseplate hinge hole of the baseplate, the first baseplate hinge hole defined by a first car of the baseplate at the open end; inserting a portion of the clamp into the slot at the open end; inserting the hinge pin through a clamp hinge hole of the clamp; and threadedly engaging the hinge pin with a second baseplate hinge hole of the baseplate, the second baseplate hinge hole defined by a second car of the baseplate at the open end, the slot extending between the first car and the second car.
(58) In some aspects, the method can further comprise tightening the hinge pin to pinch the portion of the clamp between the first car and the second ear and secure the baseplate relative to the clamp about the hinge pin. In some aspects, the baseplate can be a first baseplate, the brace assembly can be a first brace assembly, and the method can further comprise: sliding a slot of a second baseplate of a second brace assembly onto the rod through an open end of the second baseplate until the rod is positioned at a slot end of the slot of the second baseplate; and stacking the second baseplate on top of the first baseplate to interlock the second baseplate with the first baseplate and secure the second baseplate in a set orientation relative to the first baseplate about the rod.
(59) One should note that conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more particular embodiments or that one or more particular embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment.
(60) It should be emphasized that the above-described embodiments are merely possible examples of implementations, merely set forth for a clear understanding of the principles of the present disclosure. Any process descriptions or blocks in flow diagrams should be understood as representing modules, segments, or portions of code which include one or more executable instructions for implementing specific logical functions or steps in the process, and alternate implementations are included in which functions may not be included or executed at all, may be executed out of order from that shown or discussed, including substantially concurrently or in

reverse order, depending on the functionality involved, as would be understood by those reasonably skilled in the art of the present disclosure. Many variations and modifications may be made to the above-described embodiment(s) without departing substantially from the spirit and principles of the present disclosure. Further, the scope of the present disclosure is intended to cover any and all combinations and sub-combinations of all elements, features, and aspects discussed above. All such modifications and variations are intended to be included herein within the scope of the present disclosure, and all possible claims to individual aspects or combinations of elements or steps are intended to be supported by the present disclosure.

Claims

1. An assembly comprising: a first baseplate defining a first slot extending from a first open end to a first closed end, the first baseplate defining a first baseplate hinge hole at the first open end, and a first rounded end defining a first axis at the first closed end, wherein engagement between a fastener and a countersink in the first baseplate secures a rod at the first rounded end of the first baseplate; and a second baseplate defining a second slot extending from a second open end to a second closed end, the second baseplate defining a second baseplate hinge hole at the second open end, and a second rounded end defining a second axis at the second closed end; wherein the first axis is aligned with the second axis.
2. The assembly of claim 1 wherein the first baseplate comprises the countersink recessed into a first upper surface and a first lower surface opposite from the first upper surface, a first slot narrows from the first upper surface to the first lower surface of the first baseplate, and the second baseplate comprises a second upper surface, and wherein the first lower surface is stacked on top of the second upper surface.
3. The assembly of claim 1 wherein a sizing of the first baseplate cooperates with the sizing of the second baseplate to secure the rod at the first rounded end of the first baseplate.
4. The assembly of claim 3 wherein a leg of the second baseplate engages with an ear of the first baseplate to secure the rod at the first rounded end of the first baseplate.
5. An assembly comprising: a first baseplate defining a first slot extending from a first open end to a first closed end, the first baseplate defining a first baseplate hinge hole at the first open end, and a first rounded end defining a first axis at the first closed end; and a second baseplate defining a second slot extending from a second open end to a second closed end, the second baseplate defining a second baseplate hinge hole at the second open end, and a second rounded end defining a second axis at the second closed end; wherein the first baseplate comprises a countersink recessed into a first upper surface and a first lower surface opposite from the first upper surface, a first slot narrows from the first upper surface to the first lower surface of the first baseplate, and the second baseplate comprises a second upper surface, and wherein the first lower surface is stacked on top of the second upper surface; wherein the first axis is aligned with the second axis.
6. The assembly of claim 5 wherein engagement between a fastener and the countersink in the first baseplate secures a rod at the first rounded end of the first baseplate.
7. The assembly of claim 5 wherein a sizing of the first baseplate cooperates with the sizing of the second baseplate to secure a rod at the first rounded end of the first baseplate.
8. The assembly of claim 7 wherein a leg of the second baseplate engages with an ear of the first baseplate to secure the rod at the first rounded end of the first baseplate.
9. An assembly comprising: a first baseplate defining a first slot extending from a first open end to a first closed end, the first baseplate defining a first baseplate hinge hole at the first open end, and a first rounded end defining a first axis at the first closed end; and a second baseplate defining a second slot extending from a second open end to a second closed end, the second baseplate defining a second baseplate hinge hole at the second open end, and a second rounded end defining a second axis at the second closed end; wherein a sizing of the first baseplate cooperates with the

sizing of the second baseplate to secure a rod at the first rounded end of the first baseplate, and wherein the first axis is aligned with the second axis.

10. The assembly of claim 9 wherein the first baseplate comprises a countersink recessed into a first upper surface and a first lower surface opposite from the first upper surface, a first slot narrows from the first upper surface to the first lower surface of the first baseplate, and the second baseplate comprises a second upper surface, and wherein the first lower surface is stacked on top of the second upper surface.

11. The assembly of claim 9 wherein engagement between a fastener and a countersink in the first baseplate secures the rod at the first rounded end of the first baseplate.

12. The assembly of claim 9 wherein a leg of the second baseplate engages with an ear of the first baseplate to secure the rod at the first rounded end of the first baseplate.